2022-26

Full Marks : 70

Time : 3 hours

Answer from **both** the Sections as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

SECTION—A

Answer any *four* questions. 10 × 4

1. What is system analysis and design ? Explain the difference between Analysis and Design phases of software development.

2. What are the various activities involved in the feasibility study ? Discuss any three feasibilities need to be studied during feasibility analysis.

3. Why is it difficult to determine user requirement ? Illustrate any two important fact-finding techniques for getting information about user requirement.

4. What is Data Flow diagrams ? Draw context level Data flow Diagram of university admission system.

5. Explain common skills of a system analyst. Which skill do you think is most important ?

6. What is need of software documentation ? Explain any three documentation standards.

7. Define SDLC. Compare and contrast waterfall model and prototyping model.

8. Explain decision table and decision tree by taking suitable example.

SECTION—B

All questions are compulsory 3 × 10

9. How can risk and threats to IT systems be identified and managed ?

10. What are the issues and constraints to be taken care of while designing a software product ?

11. What is meant by "User Interface" ? What are the basic guidelines to design a user interface ?

12. What is meta data ?

13. What do you mean by risk analysis ?

14. What is software quality control ?

15. List any three important elements of system.

16. What are software threats ?

17. Why software testing is important, comment your views ?

18. Give example of physical and abstract system.

033

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2023(New)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. What do you mean by system ? What are the important elements of system ? Draw a neat diagram and explain them.

2. What are different classifications of system ? Explain any three by taking appropriate example.

3. Describe feasibility study ? Discuss any three important feasibility studies during software development.

4. Explain System Development Life Cycle and its various phases.

5. Draw a context level DFD of a "Hospital Management System". Make necessary assumptions.

6. What is decision table ? Design a decision table for a system of your choice, explain it.

7. What is software quality assurance ? How does it differ from software quality control.

8. What is software testing ? Why is it important ? Explain any two software testing techniques ?

Section – B

(Compulsory)

Answer **all** questions : 10×3 = 30

9. What is an information system ?

10. What is system planning ?

11. What is data dictionary ?

12. Explain briefly cost benefit analysis ?

13. What is pseudo code ?

14. What do you mean by risk analysis ?

15. What is data analysis ?

16. List three important characteristics of a good documentation.

17. Discuss briefly the importance of Software maintenance.

18. What are software threats ?



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UG — BCA
(C – 3004)

2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed

Section – A

Answer any **four** questions of the following : 10×4 = 40

1. What is the System Development Life Cycle ? How does it relate to Systems Analysis ?

2. What is software testing ? Elaborate all types of testing.

3. What are data flow diagrams ? How it is differ from structure charts ?

4. What are various types of documentation ? Discuss each type.

5. Explain Spiral model for software life cycle and write its various activities in each phase.

6. How important is a project team in feasibility analysis ? Is it mandatory in every study ?

7. Explain the advantages of top-down design.

8. What is System Implementation Process ? Discuss merits and demerits of each System Implementation Method.

Section – B

Answer **all** questions of the following : 3×10 = 30

1. What is ISO Certification 9000 ?

2. Differentiate Deterministic and Stochastic systems.

3. What software criteria are considered for selection ?

4. Give the differences between System audit and System modification.

5. Define Gantt Chart.

6. What is zero level DFD ?

7. What is the goal of input design ?

8. Give the differences between Decision tree and Decision table.

9. What levels of quality assurance must a system meet ?

10. List various characteristics of the system.

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2016-19

Full Marks : 70

Time : 3 hours

Answer from both the sections as directed

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable.

SECTION—A

Answer any *four* questions out of *eight* questions. 10×4

1. What are real time systems ? Are all online systems real time ? Explain.

2. Compare the different life cycle models of SDLC.

3. What are the various skills requirements of an analyst in the various stages of system development process ?

4. How does the size of organization affect the responsibilites of a system analyst ?

5. What is the difference between cash flow analysis and payback analysis ?

6. What is the difference between physical DFD and logical DFD ?

7. What are the different access methods of a file ?

8. Write a short note on software quality assurance.

SECTION—B

Answer *all* questions 10×3

1. Define system. What are the various type of system ?

2. What is documentation ? Explain.

3. 'A system analyst also plays the role of a psychologist'. Discuss.

4. How system planning is important for any system ?

5. What are the major design consideration of a file ?

6. What is the difference between CUI and GUI environment ?

7. Why maintenance is an important phase of SDLC ?

8. What do you mean by risk analysis ?

9. What factors will you consider while selecting software for your project ?

10. What are the features of a GUI environment ?

Risk analysis starts with planning for secure systems by identifying the vulnerability of the system and impact of this.

Risk analysis is a teamwork of experts with different backgrounds

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UGDS(III) — BCA
(2804) Sys. Anal.
& Dgn.

2018

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

Answer any **four** questions.

1. What is a system ? What are the elements of a system ? 10

2. Discuss roles and responsibilities of a system analyst. 10

3. What are the bases of planning in system analysis ? Elaborate the plainning dimensions to determine information system development. 10

4. What is system development life cycle ? In which step of SDLC, project early termination could be done ? 10

5. What do you mean by feasibility study ? Compare and contrast initial investigation and feasibility study. 10

6. Define and explain cost-benefit analysis. Discuss three categories of costs and benefits. 10

7. What are data flow diagrams ? Draw the top level DFD for library information system . 10

8. What are the characteristics of a good documentaton ? List various types of documentation. 10

Section – B

Answer **all** questions.

9. What are the characteristics of a system ? 3

10. Why is maintenance of a system necessary ? 3

11. What do you mean by data dictionary ? 3

12. What are the major threats to system security ? 3

13. Why is it difficult to determine user requirements ? 3

14. What is a decision tree ? Explain by taking example. 3

15. Differentiate between open system and closed system. 3

16. What do we need to test systems ? Elaborate 3

17. Explain briefly the different levels of quality assurance. 3

18. Explain how to do fact finding. 3

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2015-18

Full Marks : 70

Time : 3 hours

Answer both the Sections as directed

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable.

SECTION—A

Answer **four** questions out of **eight** questions

1. Discuss the various types of systems along with their characteristics. 10

2. Explain in detail the activities performed in each phase of system development life cycle. 10

3. What do you mean by feasibility study ? Describe the various types of feasibility studies. 10

4. What is cost benefit analysis ? Explain the various evaluation methods for cost benefit analysis. 10

5. What do you mean by decision table and explain the steps in creating a decision table ? Compare and contrast, structured English and decision table techniques of logic modeling. 10

6. Explain different types of system flow charts. 10

7. Explain the various quality assurance factors with example. 10

8. Explain the various types of system testing. 10

SECTION—B

All are compulsory

1. Define System. What are the elements of a system ? 3

2. What are the limitations of the waterfall model ? 3

3. Distinguish between open and closed systems. 3

4. Differentiate between Physical and Abstract System. 3

5. Differentiate between technical feasibility and operational feasibility of a project. 3

6. What is the purpose of system testing ? 3

7. What is difference between a flow chart and a data flow diagram ? 3

8. What is a decision tree ? How does it differ from decision table ? 3

9. What do you mean by risk analysis ? 3

10. What are different types of threats to Computer System ? 3